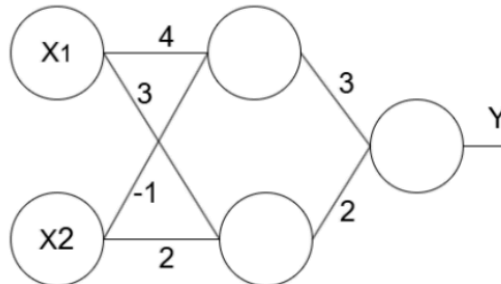


BRAC UNIVERSITY
Department of Computer Science and Engineering
CSE 422: Artificial Intelligence
MLP Problem Set

Question-01



- If $X_1=5$ and $X_2=1$, apply forward propagation to calculate the value of Y . Consider that both the hidden layer and output layer is using sigmoid as the activation function
- Mention the importance of activation functions in neural networks
- Write the mathematical expression of the gradients for the backward propagation of the neurons in the hidden layer (As demonstrated in class)

Question-02

Consider a single neuron (perceptron) with one input x and one output y

Initial weight: $w=0.4$ Initial bias: $b=0.1$ Learning rate: $\eta=0.1$

Perform one single iteration showing forward pass, weight update and back propagation

Question-03

In this part we train a feedforward neural network with one feature.

Network Structure:

- **1 input feature**, Input: $x_1 = 1$
 - **Hidden Layer 1**: 2 neurons
 - **Hidden Layer 2**: 1 neurons
 - **Output Layer**: 1 neuron
 - Activation: sigmoid at all layers $\sigma(z) = 1 / (1 + e^{-z})$
- Based on the scenario above, draw a schematic diagram of the feedforward neural network. Clearly label: each layer and its neurons, the outputs (activations) of the neurons, the flow of data through the network. [1]
 - The activation outputs from Hidden Layer 1 are 0.4 and 0.7. The weights for Hidden Layer 2 are 0.1 and 0.2. What is the activation output from the second layer? [3]